

● MEDIUM

● ADVANCED MATH

● ANSWER KEY

Nonlinear functions

06

10 answers · Rationale-rich · Teach from doc

Q1**D**

D. $gx = 7x^3 - 2$

Choice D is correct. If the graph of $y = gx$ is the result of shifting the graph of $y = fx$ down k units in the xy -plane, the function g can be defined by an equation of the form $gx = fx - k$. It's given that $fx = 7x^3$ and the graph of $y = gx$ is the result of shifting the graph of $y = fx$ down 2 units. Substituting $7x^3$ for fx and 2 for k in the equation $gx = fx - k$ yields $gx = 7x^3 - 2$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect. This equation defines a function g for which the graph of $y = gx$ is the result of shifting the graph of $y = fx$ up, not down, 2 units.

Q2**B**

B. $C = 600(0.85)^t$

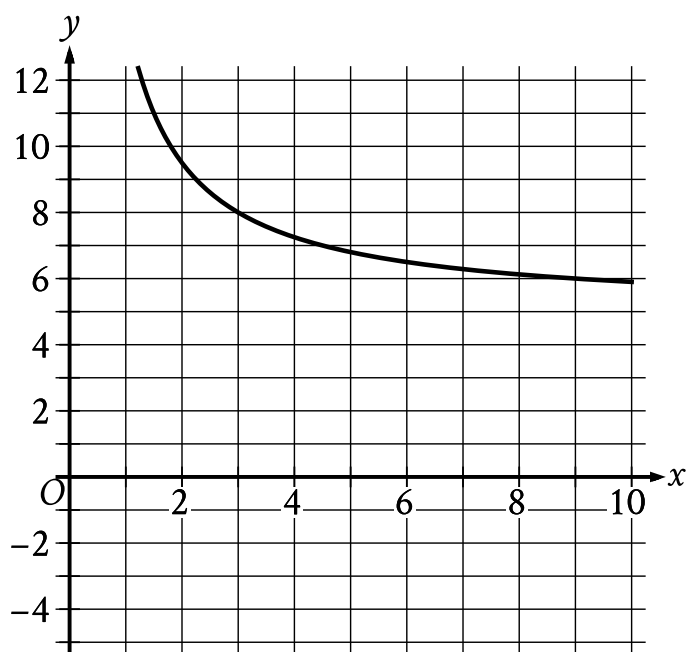
Choice B is correct. A model for a quantity C that decreases by a certain percentage per time period t is an exponential equation in the form $C = I\left(1 - \frac{r}{100}\right)^t$, where I is the initial value at time $t = 0$ for $r\%$ annual decline. It's given that C is the number of millions of CDs sold in the United States and that t is the number of years after 2005. It's also given that 600 million CDs were sold at time $t = 0$, so $I = 600$. This number declines by 15% per year, so $r = 15$.

Substituting these values into the equation produces $C = 600\left(1 - \frac{15}{100}\right)^t$, or $C = 600(0.85)^t$.

Choice A is incorrect and may result from errors made when representing the percent decline. Choices C and D are incorrect. These equations model exponential increases in CD sales, not exponential decreases.

Q3

D



D.

- Moving from left to right:
 - The curve is shown in quadrant 1.
 - The curve trends down sharply to x equals 3.
 - The curve then trends down gradually.
- As x increases, the curve approaches the line y equals 5.
- The curve passes through the following points:
 - $(2, \frac{19}{2})$
 - $(3, 8)$
 - $(6, \frac{13}{2})$

Choice D is correct. It's given that the graph of the rational function f is shown, where $y = fx$ and $x \geq 0$. The graph shown passes through the point 3, 3. It follows that when the value of x is 3, the value of fx is 3. When the value of fx is 3, the value of $fx + 5$ is $3 + 5$, or 8. Therefore, the graph of $y = fx + 5$ passes through the point 3, 8. Of the given choices, choice D is the only graph that passes through the point 3, 8 and is therefore the graph of $y = fx + 5$.

Choice A is incorrect. This is the graph of $y = fx - 5$, rather than $y = fx + 5$.

Choice B is incorrect. This is the graph of $y = \frac{fx}{5}$, rather than $y = fx + 5$.

Choice C is incorrect and may result from conceptual or calculation errors.

Q4

B

B. 790

Choice B is correct. It's given that t minutes after an initial observation, the number of bacteria in a population is $40,0002^{\frac{t}{790}}$. This expression consists of the initial number of bacteria, 40,000, multiplied by the expression $2^{\frac{t}{790}}$. The time, in minutes, it takes for the number of bacteria to double is the increase in the value of t that causes the expression $2^{\frac{t}{790}}$ to double. Since the base is 2, the expression $2^{\frac{t}{790}}$ will double when the exponent increases by 1. Since the exponent of this expression is $\frac{t}{790}$, the exponent will increase by 1 when t increases by 790. Therefore, the time, in minutes, it takes for the number of bacteria in the population to double is 790.

Choice A is incorrect. This is the base of the exponent, not the time it takes for the number of bacteria in the population to double.

Choice C is incorrect. This is the number of minutes it takes for the population to double twice.

Choice D is incorrect. This is the number of bacteria that are initially observed, not the time it takes for the number of bacteria in the population to double.

Q5

B

B. $f(x) = (x - 24)(x - 30) - 8$

Choice B is correct. According to the table, $f(24) = -8$, $f(30) = -8$, and $f(32) = 8$. Let the quadratic function g be defined by $g(x) = f(x) + 8$. Therefore, $g(24) = f(24) + 8$, $g(30) = f(30) + 8$, and $g(32) = f(32) + 8$. Substituting -8 for $f(24)$ in the equation $g(24) = f(24) + 8$ yields $g(24) = -8 + 8$, or $g(24) = 0$. Substituting -8 for $f(30)$ in the equation $g(30) = f(30) + 8$ yields $g(30) = -8 + 8$, or $g(30) = 0$. Substituting 8 for $f(32)$ in the equation $g(32) = f(32) + 8$ yields $g(32) = 8 + 8$, or $g(32) = 16$. By the factor theorem, if $g(24) = 0$ and $g(30) = 0$, then $g(x) = k(x - 24)(x - 30)$, where k is a constant. Since $g(32) = 16$, substituting 32 for x and 16 for $g(x)$ in the equation $g(x) = k(x - 24)(x - 30)$ yields $16 = k(32 - 24)(32 - 30)$, or $16 = k(8)(2)$, which is equivalent to $16 = 16k$. Dividing both sides of this equation by 16 yields $1 = k$. Substituting 1 for k in the equation $g(x) = k(x - 24)(x - 30)$ yields $g(x) = (x - 24)(x - 30)$. Since $g(x) = f(x) + 8$, it follows that $g(x) - 8 = f(x)$. Subtracting 8 from both sides of the equation $g(x) = (x - 24)(x - 30)$ yields $g(x) - 8 = (x - 24)(x - 30) - 8$. Substituting $f(x)$ for $g(x) - 8$ in this equation yields $f(x) = (x - 24)(x - 30) - 8$.

Choice A is incorrect. If this equation defined f , then when $x = 24$, the value of $f(x)$ would be 4 , not -8 .

Choice C is incorrect. If this equation defined f , then when $x = 32$, the value of $f(x)$ would be 32 , not 8 .

Choice D is incorrect. If this equation defined f , then when $x = 32$, the value of $f(x)$ would be -32 , not 8 .

Q6**A**

A. The estimated number of employees when the restaurant opened

Choice A is correct. For an exponential function of the form $E_t = ab^t$, where a and b are constants, the initial value of the function—that is, the value of the function when $t = 0$ —is a and the value of the function increases by a factor of b each time t increases by 1. Since the function $E_t = 51.8^t$ gives the estimated number of employees at a restaurant and t is the number of years since the restaurant opened, the best interpretation of the number 5 in this context is the estimated number of employees when $t = 0$, or when the restaurant opened.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Q7**D**

D. 8 years after Immanuel purchased the rare coin, its predicted value is approximately 82 dollars.

Choice D is correct. It's given that the function $f_x = 651.03^x$ gives the predicted value, in dollars, of a certain rare coin x years after Immanuel purchased it. It follows that f_x represents the predicted value, in dollars, of the coin x years after Immanuel purchased it. Since the value of f_8 is the value of f_x when $x = 8$, it follows that “ f_8 is approximately equal to 82” means that f_x is approximately equal to 82 when $x = 8$. Therefore, the best interpretation of the statement “ f_8 is approximately equal to 82” in this context is 8 years after Immanuel purchased the rare coin, its predicted value is approximately 82 dollars.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Q8

A

A. The length of the rectangle, in feet

Choice A is correct. It's given that a rectangle has a length that is 15 times its width. It's also given that the function $y = 15ww$ represents this situation, where y is the area, in square feet, of the rectangle and $y > 0$. The area of a rectangle can be calculated by multiplying the rectangle's length by its width. Since the rectangle has a length that is 15 times its width, it follows that w represents the width of the rectangle, in feet, and $15w$ represents the length of the rectangle, in feet. Therefore, the best interpretation of $15w$ in this context is that it's the length of the rectangle, in feet.

Choice B is incorrect. This is the best interpretation of y , not $15w$, in the given function.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect. This is the best interpretation of w , not $15w$, in the given function.

Q9**D**

D. $p = 15,000(1.04)^x$

Choice D is correct. It's given that a model predicts the population of Bergen in 2005 was 15,000. The model also predicts that each year for the next 5 years, the population increased by 4% of the previous year's population. The predicted population in one of these years can be found by multiplying the predicted population from the previous year by 1.04. Since the predicted population in 2005 was 15,000, the predicted population 1 year later is $15,000(1.04)$. The predicted population 2 years later is this value times 1.04, which is $15,000(1.04)^2$, or $15,000(1.04)^2$. The predicted population 3 years later is this value times 1.04, or $15,000(1.04)^3$. More generally, the predicted population, p , x years after 2005 is represented by the equation $p = 15,000(1.04)^x$.

Choice A is incorrect. Substituting 0 for x in this equation indicates the predicted population in 2005 was 0.96 rather than 15,000.

Choice B is incorrect. Substituting 0 for x in this equation indicates the predicted population in 2005 was 1.04 rather than 15,000.

Choice C is incorrect. This equation indicates the predicted population is decreasing, rather than increasing, by 4% each year.

Q10**18**

The correct answer is 18. It's given that an egg is thrown from a rooftop and that the equation $h = -4.9t^2 + 9t + 18$ represents this situation, where h is the height of the egg above the ground, in meters, t seconds after it is thrown. It follows that the height, in meters, from which the egg was thrown is the value of h when $t = 0$. Substituting 0 for t in the equation $h = -4.9t^2 + 9t + 18$ yields $h_0 = -4.9(0)^2 + 9(0) + 18$, or $h = 18$. Therefore, according to the equation, the height, in meters, from which the egg was thrown is 18.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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